A Appendix

A.1 Dataset Creation Details

In section , we discussed the source data collection process of DRIFTBENCH. We outlines the key implementation details and the prompt templates used to generate the diversified and fabricated content in the DRIFTBENCH, including both text and image transformations.

Text diversification (*DT*) : Given a human written news text (*OT*), we prompt GPT-4o (Hurst et al. 2024) to rephrase the original text while preserving its semantics.

> You are given a news caption. Your task is to rewrite it while ensuring the original meaning remains unchanged.
> News caption: {}
> Your Response:

Text fabrications (*FT*) : Given a human written news text (*OT*), we prompt GPT-4o (Hurst et al. 2024) to generate a false narrative by altering the original news caption.

> You are given a news caption. Your task is to rewrite the event described in the original caption with a false claim.
> News caption: {}
> Your Response:

Image diversification (*DI*) : Given an authentic image (*OI*), we utilize *FLUX.1 Redux [dev]* (Labs et al. 2025), an open-source image-to-image variant generation model, to generate diversified image. We set the *guidance_scale* as 5.5 and *num_inference_step* as 50.

Image fabrications (*FI*) : Given an false narrative (*FI*), we utilize *FLUX.1 [dev]* (Labs 2024), an open-source text-to-image generation model, to generate fabricated image. We set the *guidance_scale* as 3.5 and *num_inference_step* as 50.

> Generating an image based on the text description in a natural style. Text Description: [FT]

A.2 Human Evaluation

We illustrate the annotation process for human evaluation discussed in Section in Figure 7.

A.3 Malicious Evidence Contamination

To implement malicious evidence contamination through guided retrieval, we prompt GPT-4o (Hurst et al. 2024) to generate evidence with malicious intent, such as evidence supporting or refuting news instances, and directly insert it into the original evidence list. The original evidence list is curated by crawling data from the web and selecting the top 5 pieces of evidence. To evaluate the impact of evidence contamination, we insert the generated malicious evidence into the list and use it for model inference.

> You are given an news image and corresponding news caption. Your task is to rewrite this description into a **supporting/refuting** event — that is, to **support/refute** the original caption.
> Original caption: {}
> Your response:

A.4 Error Attribution

We listed detailed error attribution taxonomy in Figure 8 discussed in Section .

Error attribution identifies two types of errors caused by two levels of drift: model misperception drift errors and evidence drift errors. Model misperception drift errors are directly caused by diversified input, leading to model misclassification, and are unrelated to external evidence. Evidence drift errors occur when the model's misclassification is triggered by changes in the external retrieved evidence following diversification.

A.5 Explanation Evaluation

To analyze the reasoning behavior of the model, we evaluated the explanations generated by different methods discussed in . The detailed evaluation dimensions and criteria is shown in Figure 9, and the results for all models are shown in the Figure 10. In this part, we also use GPT-4o (Hurst et al. 2024) to conduct our experiment.

A.6 Detailed Results

We present the detailed performance of *Real vs. Fake* instances across all diversified types discussed in Section in Table 4. Additionally, we report the results of robustness to evidence contamination in Table 5, as discussed in Section .

A.7 Evaluation Prompts

We present our prompts for evaluating Vanilla LVLMs in Figure 11 .

Figure 7: llustration of human evaluation instructions for validating the quality of DriftBench. We present the validation process of *DI_DT*, with the validation processes for *DI_OT* and *OI_DT* being similar.

Model misperception drift errors:

- **Text–Image inconsistency**: The image and text do not align, leading to contradictions, such as the wrong person, place, action, or object being depicted.
- **Text–Image ambiguous or weak match**: The image is too vague or the caption is too generic, making it difficult to establish a clear and meaningful connection between them.
- **Image and caption match, but the claim itself is fabricated**: The image and caption are consistent, but the event or claim described is fabricated or never occurred (e.g., a historical event that never happened).

Model misperception drift errors:

- **No supporting evidence found**: The retrieval process fails to find any relevant evidence or retrieves evidence from irrelevant domains, leading to a lack of support for the claim.
- **Misleading retrieved evidence**: The retrieved evidence may have similar names or topics to the claim but is ultimately a poor or incorrect match.
- **Mismatch in time or location between the evidence and the caption**: The time or location of the retrieved evidence does not align with that presented in the caption, leading to inconsistencies.
- **Over-reliance on low-quality or noisy evidence**: The model excessively depends on low-quality or noisy evidence, which undermines the overall reliability and validity of the claim's verification.

Figure 8: Error attribution taxonomy according to model-level misperception drift and evidence-level drift.

- **Justification Validity (1–5):** Based on the Ground Truth Label, how well does explanation support the correct label?
  - Score 5: Justification is completely sound and logical.
  - Score 1: Justification is flawed or supports the wrong conclusion.
- **Evidence Grounding (1–5):** Does explanation explicitly and accurately attribute its claims to the specific (drifted) evidence?
  - Score 5: Each key argument is precisely linked to a specific piece of evidence.
  - Score 1: Makes claims without any reference to the evidence.
- **Evidence Synthesis (1–5):** Does explanation weave multiple points from evidence into a coherent argument?
  - Score 5: Skillfully combines multiple evidence points to form a robust conclusion.
  - Score 1: Fails to handle multiple evidence points, leading to confusion.
- **Reasoning Depth (1–5):** Does the reasoning go beyond surface-level matching?
  - Score 5: Uncovers deep, underlying consistencies or inconsistencies.
  - Score 1: Reasoning is superficial and based on simple keyword matching.
- **Debunking Power (1–5):** How effective would explanation be at convincing a layperson of the ground truth?
  - Score 5: Highly persuasive, clear, and provides helpful context.
  - Score 1: Vague, unconfident, or too brief to be persuasive.

Figure 9: Five evaluation dimensions for explanations generated by MMD method.

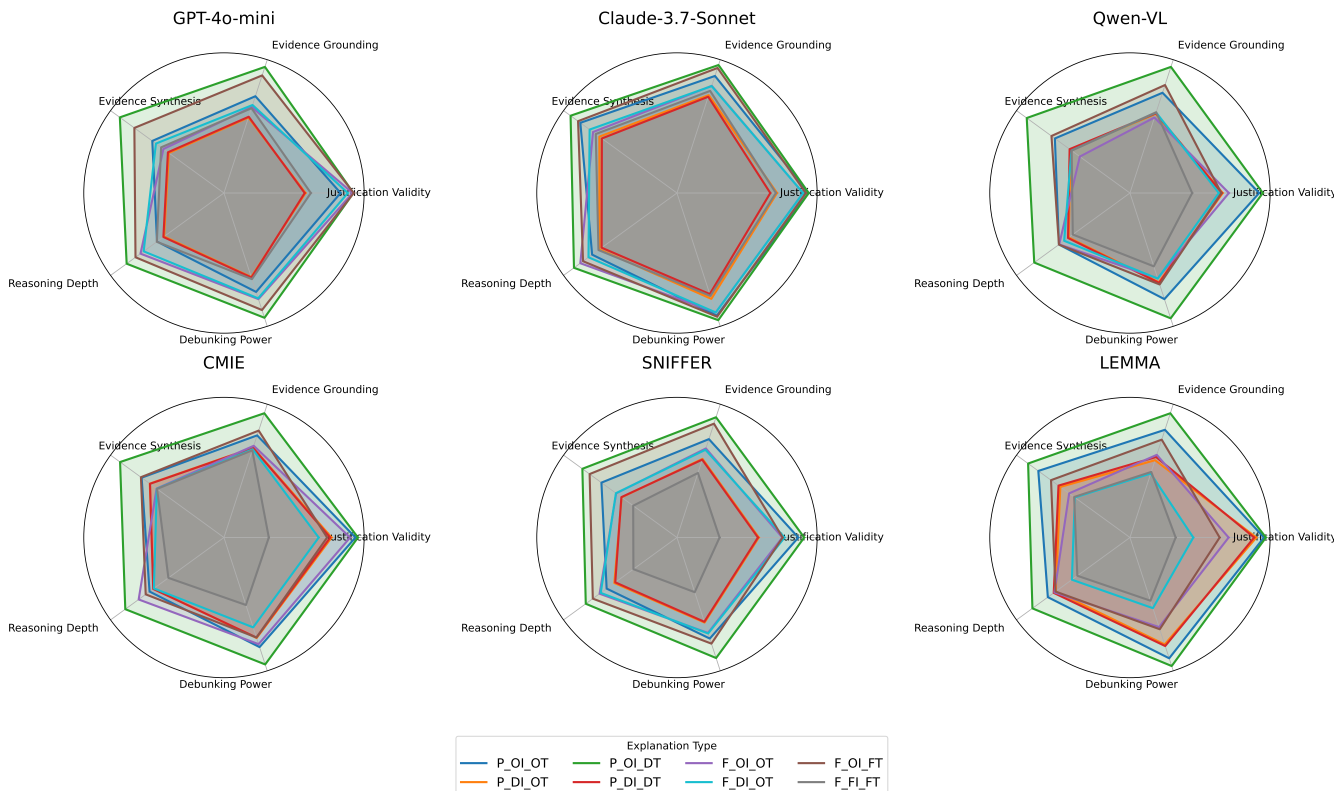

Figure 10: Model reasoning capabilities under different dimension.

Table 4: Detailed comparison of LVLM-based MMD methods under the all types diversification.

Infer Type	Data Type	Real		Fake	
		Recall	F1	Recall	F1
GPT-4o-mini	OI_OT	75.6	86.1	91.1	95.3
	DI_OT	35.6 (-40.0)	52.6 (-33.5)	93.1 (+2.0)	96.4 (+1.1)
	OI_DT	74.7 (-0.9)	85.5 (-0.6)	94.5 (+3.4)	97.1 (+1.8)
	DI_DT	34.7 (-40.9)	51.5 (-34.6)	64.2 (-26.9)	78.2 (-17.1)
Claude-3.7-Sonnet	OI_OT	89.2	94.3	87.5	93.3
	DI_OT	53.5 (-35.7)	69.7 (-24.6)	91.4 (-35.7)	95.5 (-24.6)
	OI_DT	83.7 (-5.5)	91.2 (-3.1)	94.5 (-5.5)	97.1 (-3.1)
	DI_DT	45.4 (-43.8)	62.5 (-31.8)	75.5 (-43.8)	86.1 (-31.8)
Qwen-VL	OI_OT	91.1	95.3	56.3	72.1
	DI_OT	62.1 (-28.4)	77.1 (-18.2)	64.7 (-78.6)	78.6 (-24.6)
	OI_DT	89.2 (-1.9)	94.3 (-1.0)	59.8 (-5.5)	74.8 (-3.1)
	DI_DT	61.9 (-29.2)	74.5 (-20.8)	45.7 (-43.8)	62.8 (-31.8)
CMIE	OI_OT	93.6	96.7	88.2	93.7
	DI_OT	72.8 (-20.8)	84.2 (-12.5)	72.1 (-16.1)	83.8 (-9.9)
	OI_DT	93.7 (+0.1)	96.8 (+0.1)	75.1 (-13.1)	85.5 (-8.2)
	DI_DT	72.4 (-21.2)	83.9 (-12.8)	23.1 (-65.1)	37.5 (-57.8)
SNIFFER	OI_OT	93.2	96.5	76.2	86.5
	DI_OT	76.0 (-17.2)	86.4 (-10.1)	71.1 (-5.1)	83.1 (-3.4)
	OI_DT	93.4 (+0.2)	96.6 (+0.1)	78.3 (+2.1)	87.8 (+1.3)
	DI_DT	74.8 (-18.4)	85.6 (-10.9)	17.9 (-0.0)	30.3 (-0.0)
LEMMA	OI_OT	92.8	96.2	77.5	79.8
	DI_OT	83.1 (-9.7)	90.7 (-5.5)	54.7 (-22.8)	70.7 (-9.1)
	OI_DT	92.0 (-0.8)	95.8 (-0.4)	73.8 (-3.7)	84.9 (+5.1)
	DI_DT	82.8 (-10.0)	90.6 (-5.6)	25.5 (-65.6)	40.6 (-54.7)

Table 5: Detailed comparison of LVLMs-based MMD performance with malicious evidence contamination under the *DI_OT* controlled news diversity.

Infer Type	Data Type	Accuracy	Real			Fake		
			Precision	Recall	F1	Precision	Recall	F1
GPT-4o-mini	DI_OT	64.4	83.7	35.7	50.0	59.1	93.1	72.3
	Polluted	44.6 (-19.8)	18.8 (-64.9)	3.3 (-32.4)	5.5 (-44.5)	47.1 (-12.0)	86.0 (-7.1)	60.8 (-11.5)
Claude-3.7-Sonnet	DI_OT	72.4	86.1	53.5	66.0	66.3	91.4	76.8
	Polluted	66.7 (-5.7)	79.0 (-7.1)	45.4 (-8.1)	57.7 (-8.3)	61.7 (-4.6)	87.9 (-3.5)	72.5 (-4.3)
Qwen-VL	DI_OT	63.7	63.9	62.7	63.3	63.5	64.7	64.1
	Polluted	37.2 (-26.5)	32.3 (-31.6)	23.2 (-39.5)	27.0 (-36.3)	39.9 (-23.6)	51.2 (-13.5)	44.8 (-19.3)
CMIE	DI_OT	72.4	72.2	72.8	72.5	72.6	72.1	72.3
	Polluted	66.8 (-5.6)	66.9 (-5.3)	66.2 (-6.6)	66.6 (-5.9)	66.6 (-6.0)	67.3(-4.8)	66.9 (-5.4)
SNIFFER	DI_OT	73.5	71.2	76.0	73.5	75.8	71.1	73.4
	Polluted	70.3 (-3.2)	70.2 (-1.0)	73.4 (-2.6)	71.8 (-1.7)	70.4 (-5.4)	67.1(-4.0)	68.7 (-4.7)
LEMMA	DI_OT	68.9	64.7	83.1	72.7	76.3	54.7	63.7
	Polluted	41.9 (-27.0)	42.1 (-22.6)	43.2 (-39.9)	42.7 (-30.0)	41.7 (-34.6)	40.7 (-14.0)	41.2 (-22.5)

Misinformation Detection Using Vanilla LVLMs

You are given a news post, inncluding an image, a piece of text, and a list of titles in which the image has appeared. Your task is to predict whether there is misinformation between the given image and text.
Generate a JSON object with two properties: 'label', 'explanation'. The return value of 'label' property should be selected from ["real", "fake"]. Fake indicates there is misinformation between the given image and text. Real indicates that there is no misinformation between the given image and text. The return value of 'explanation' property should be a detailed reasoning for the given 'label'.
The given text: {}
The list of titles related to the image content: {}
Your Response:

Figure 11: Prompt of misinformation detection using Vanilla LVLMs.